

Giuseppe Durisi: Curriculum Vitae

Work address:

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Chalmers University of Technology
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RESEARCH	Information theory, communication theory, machine learning
PERSONAL	Born on November 28, 1977 in Turin, Italy; Italian citizenship; male; married with two children; languages: Italian, English, German, Swedish
EDUCATION	◇ Mar. 2012: Chalmers University of Technology, Sweden Docent diploma in Information Theory ◇ Mar. 2006: Politecnico di Torino, Italy Ph.D. in electrical and communication engineering · Doctoral dissertation: <i>Noncoherent fading channels: capacity and coding</i> · Advisor: Professor Sergio Benedetto (Politecnico di Torino, Italy) ◇ Nov. 2001: Politecnico di Torino, Italy Telecommunications Engineering Diploma [M.Sc.], <i>summa cum laude</i> · Diploma dissertation: <i>Analysis and simulation of an ultra-wideband communication system over a multipath channel</i> · Advisors: Professor Sergio Benedetto (Politecnico di Torino, Italy), Dr. Giovanni Romano (Telecom Italia Lab, Turin, Italy)
ACADEMIC WORK EXPERIENCE	◇ Jun. 2020–current: Full professor at Chalmers University of Technology, Sweden ◇ Apr. 2023–current: Director of the master program in Information and Communication Technologies at Chalmers University of Technology, Sweden ◇ Feb. 2016–Feb. 2022: Co-director Information and Communication Technology (ICT) Area of Advance at Chalmers University of Technology, Sweden ◇ Oct. 2018–Dec. 2020: Co-director of Chalmers Artificial Intelligence Research Center (CHAIR) ◇ Jan. 2017–May 2020: Professor at Chalmers University of Technology, Sweden ◇ Dec. 2013–Dec. 2016: Associate professor at Chalmers University of Technology, Sweden ◇ Nov. 2010–Nov. 2013: Assistant professor at Chalmers University of Technology, Sweden ◇ Apr. 2012: Guest professor at Vienna University of Technology, Austria ◇ Feb. 2008–Oct. 2010: Senior researcher at ETH Zurich, Switzerland ◇ Feb. 2006–Jan. 2008: Postdoctoral researcher at ETH Zurich, Switzerland ◇ Nov. 2004–Nov. 2005: Visiting researcher at ETH Zurich, Switzerland ◇ Mar. 2004–Jul. 2004: Visiting researcher at Università di Pisa, Italy ◇ Jan. 2002–Jan. 2006: Junior researcher at Istituto Superiore Mario Boella, Turin, Italy
INDUSTRIAL WORK EXPERIENCE	◇ Sep. 2018–Sep. 2020: Visiting researcher at Qamcom Research & Technology AB, Sweden ◇ Jan. 2014–2017: Visiting researcher at Ericsson Research, Sweden ◇ Nov. 2002–Apr. 2003: Visiting researcher at IMST GmbH, Germany ◇ Feb. 2001–Nov. 2001: Master thesis internship at Telecom Italia Lab, Turin, Italy

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- HONORS
- ◇ 2025: *best paper* at the International Symposium on Information Theory Workshop "Learn to Compress or Compress to Learn"
 - ◇ 2022: *3rd place award* at the Asilomar Conference on Signals, Systems and Computers *student paper contest* (lead author: Ph.D. student Oguz Kislal)
 - ◇ 2017: exemplary reviewer, IEEE Transactions on Wireless Communications
 - ◇ 2016: exemplary reviewer, IEEE Transactions on Communications
 - ◇ 2013 IEEE ComSoc Young Researcher Award for Europe, Middle East, and Africa Region
 - ◇ 2013 IEEE Sweden VT-COM-IT Chapter *best student conference paper award* (lead author: Ph.D. student Wei Yang)
 - ◇ *Student paper award* at the IEEE International Symposium on Information Theory (ISIT 2012) (lead author: Ph.D. student Wei Yang)
 - ◇ *Senior member* of the Institute of Electrical and Electronics Engineers (IEEE), Feb. 2012
 - ◇ *Optime* award from Unione Industriale, Turin, Italy (for best graduates of the year 2001)
- TEACHING & SUPERVISION
- ◇ Chalmers University of Technology: **Lecturer**
 - 2021–2025: Probability and statistical learning using python (7.5 credits, master course, fall semester, taught in English)
 - 2020–2025: Statistics and machine learning in high dimensions (7.5 credits, master/PhD course, fall semester, taught in English)
 - 2011–2025: Information theory (7.5 credits, master/PhD course, spring semester, taught in English)
 - 2012–2023: Wireless communications (7.5 credits, master course, spring semester, taught in English, *guest lecturer*)
 - 2021–2022: Digital communications (7.5 credits, master course, fall semester, taught in English)
 - 2016–2018: Probability and random processes (7.5 credits, master/PhD course, fall semester, taught in English)
 - 2014–2018: Electrical measurements: systems and methods (7.5 credits, bachelor course, spring semester, taught in English/Swedish)
 - 2011: Network information theory (7.5 credits, PhD course, fall semester, taught in English)
 - ◇ Vienna University of Technology: **Lecturer**
 - 2012: Network information theory (1.5 unit, graduate/PhD short course, spring semester, taught in English)
 - ◇ ETH Zurich: **Teaching assistant** for graduate courses
 - 2007–2010: Fundamentals of wireless communications (4-unit graduate course, spring semester, taught in English)
 - 2009: Receivers for wireless communications (4-unit graduate course, spring semester, taught in English)
 - 2009: Harmonic analysis: theory and applications in advanced signal processing (4-unit graduate course, spring semester, taught in English)
 - ◇ Politecnico di Torino: **Teaching assistant** for undergraduate courses
 - 2005: Analysis of discrete-time signals (2-unit undergraduate course, fall semester, taught in Italian)
 - 2003–2004: Signal theory (2-unit undergraduate course, fall semester, taught in Italian)
 - ◇ Currently supervised Ph.D. students at Chalmers University of Technology:

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- Z. Cheong, *Compression and discrimination in modern machine-learning algorithms*, started in 2025
- E. Ametsi, *Runtime safe monitors for automotive ML-base functions*, started in 2025 (co-supervisor)
- A. V. Parambath, *Distributed radio networks*, started in 2024 (co-supervisor)
- B. Chen, *Machine learning over communication systems*, started in 2024 (co-supervisor)
- K. Okumus, *Goal-oriented communications*, started in 2023 (co-supervisor)
- A. Bordbar, *Machine learning for communications*, started in 2023
- J. Naram, *Information theoretic generalization bounds*, started in 2023
- ◇ Current post-doctoral fellows at Chalmers University of Technology:
 - Deekshith P. K., *Uncoordinated massive random access*, 2023–present
- ◇ Past post-doctoral fellows at Chalmers University of Technology:
 - U. K. Ganesan, *Machine learning over communication systems*, 2024–2025
 - K-H. Ngo, *Secure and private massive cloud/edge connectivity*, 2020–2024
 - A. Lancho, *Massive multiple-access communications for IoT applications*, 2019–2021
 - A. Rezazadeh, *Information-theoretic methods in machine learning*, 2019–2022
 - G. C. Ferrante, *Finite blocklength information theory*, 2017–2018
- ◇ Past Ph.D. students
 - L. Aabel, *Distributed Antenna Systems Using 1-bit Radio-Over-Fiber*, PhD 2025 (main supervisor, Chalmers University of Technology)
 - Y. Ettefagh, *Positioning, Synchronization, and Communication in 6G: Insights from Massive MIMO, RIS, and NTN*, PhD 2025 (co-supervisor, Chalmers University of Technology)
 - O. Kislal, *Fundamental Limits of Ultra-Reliable Low-Latency Communications*, PhD 2024 (co-supervisor, Chalmers University of Technology)
 - F. Hellström, *Information-Theoretic Generalization Bounds: Tightness and Expressiveness*, PhD 2023 (main supervisor, Chalmers University of Technology)
 - J. Östman, *Ultra-Reliable Short-Packet Communications: Fundamental Limits and Enabling Technologies*, PhD 2020 (co-supervisor, Chalmers University of Technology)
 - S. Jacobsson, *Massive multi-antenna communications with low-resolution converters*, PhD 2019 (main supervisor, Chalmers University of Technology)
 - K. Keykhosravi, *Capacity analysis and receiver design in the presence of fiber nonlinearity*, PhD 2019 (co-supervisor, Chalmers University of Technology)
 - R. Devassy, *Delay and throughput bounds for the design of next generation wireless systems*, PhD 2019 (main supervisor, Chalmers University of Technology)
 - W. Yang, *Fading channels: capacity and channel coding rate in the finite-blocklength regime*, PhD 2015 (main supervisor, Chalmers University of Technology)
 - M. R. Khanzadi, *Phase noise in communication systems: modeling, compensation, and performance analysis*, PhD 2015 (co-supervisor, Chalmers University of Technology)
 - P. Kuppinger, *General uncertainty relations and sparse signal recovery*, PhD 2011 (co-supervisor, ETH Zurich)
 - V. I. Morgenthern, *Crystallization and noncoherence in wireless communication*, PhD 2010 (co-supervisor, ETH Zurich)
 - U. G. Schuster, *Wireless communication over wideband channels*, PhD 2007 (co-supervisor, ETH Zurich)

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PROFESSIONAL SERVICE

- ◊ Editorships:
 - Associate editor, IEEE Transactions on Information Theory, 2024–current
 - Guest editor, IEEE Journal on Selected Areas in Information Theory; Special issue on “Information Theoretic Foundations of Future Communication Systems”, 2021
 - Associate editor, IEEE Transactions on Communications, 2015–2021
 - Publications editor, IEEE Transactions on Information Theory, 2011–2014.
- ◊ Member of the organizing committee:
 - Tutorial chair, International Symposium on Information Theory (ISIT), Sorrento, Italy, June 2027
 - Track chair, European Conference on Networks and Communications (EUCNC), Gothenburg, Sweden, June 2023
 - Recent-result chair, International Symposium on Information Theory (ISIT), Espoo, Finland, June 2022
 - Publication chair, Information Theory Workshop (ITW), Riva del Garda, Italy, September 2020
 - Track chair, Asilomar Conference on Signals, Systems, and Computers, Pacific Grove, California, USA, Nov. 2019
 - Publication chair, International Symposium on Information Theory (ISIT), Aachen, Germany, Jul. 2017
 - Chair, European School of Information Theory (ESIT), Gothenburg, Sweden, May 2016
 - Local organizer, 7th *International Symposium on Turbo Codes & Iterative Information Processing (ISTC)*, Gothenburg, Sweden, Aug. 2012
 - Chair, Workshop on *Machine Learning and Large Data Sets*, Gothenburg, Sweden, Aug. 2011
 - Chair, 1st workshop on *Ultra-Wideband for Wireless Personal Area and Sensor Networks*, Turin, Italy, Nov. 2004
- ◊ Technical program committee membership:

<i>IEEE Int. Conf. Commun. (ICC):</i>	2006, 2011–2016, 2018–2026
<i>IEEE MeditCom:</i>	2026
<i>IEEE Global Commun. Conf. (GLOBECOM):</i>	2011–2013, 2018, 2021–2026
<i>Int. Workshop on Smart Antennas (WSA):</i>	2024
<i>IEEE Int. Workshop Signal Process. Advances Wireless Commun. (SPAWC):</i>	2021–2024
<i>Int. Zurich Seminar (IZS):</i>	2014–2024
<i>IEEE Int. Conf. Comp. Comm. (INFOCOM):</i>	2023
<i>Int. Veh. Tech. Conf. (VTC):</i>	2017–2019
<i>Int. Conf. Telecommun. (ICT)</i>	2015, 2018
<i>Int. Symp. Wirel. Commun. Sys. (ISWCS):</i>	2011, 2017, 2018
<i>Int. ITG Conf. Sys. Commun. Coding (SCC):</i>	2017, 2019, 2022
<i>IEEE Inf. Theory. Workshop (ITW):</i>	2016
<i>IEEE Wirel. Commun. Net. Conf. (WCNC):</i>	2013–2016, 2019
<i>Int. Workshop Physical Wirel. Comm. Net. (PHYSCOMNET):</i>	2014
<i>Int. Conf. Comput. Net. Commun. (ICNC):</i>	2012, 2018
<i>Int. Symp. Personal, Indoor, Mobile Radio Commun. (PIMRC):</i>	2012
<i>IEEE Swedish Commun. Tech. Workshop (SWE-CTW):</i>	2011, 2012
<i>Int. Conf. on Commun. Net. in China (CHINACOM):</i>	2010
<i>IEEE Int. Conf. Ultra-Wideband (ICUWB):</i>	2009

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◇ Invited-session organizer at

<i>Asilomar Conference on Signals, Systems, and Computers:</i>	2022
<i>IEEE Inf. Theory Workshop (ITW):</i>	2020
<i>Int. Zurich Seminar (IZS):</i>	2016
<i>Int. Symp. Wirel. Commun. Sys. (ISWCS):</i>	2014

◇ Referee for the European Research Council (ERC)

◇ Referee for the Swedish Research Council (Vetenskapsrådet)

◇ Referee for the Israel Science Foundation

◇ Referee for the Italian agency for the evaluation of universities and research institutes (AN-VUR)

◇ Referee for the German Verband der Elektrotechnik Elektronik Informationstechnik (VDE-ITG)

◇ Referee for the Austrian Wiener Wissenschaft- Forschungs- und Technologiefonds (WWTF)

◇ Member of examination committee/opponent in research degree examinations

- Xinying Zou, *Generalization Theory and Algorithm Robustification in Statistical Learning*, Ph.D. Thesis, Centre Inria d'Université Côte D'Azur, France, Feb. 2026.
- Patrick Agostini, *Grant-Free Random Access for Massive Access: Coding, Detection, and Semantic-Aware Techniques*, Ph.D. Thesis, Berlin, Germany, Dec. 2025.
- Jianan Bai, *Enabling Technologies for Latency-Critical Wireless Systems*, Ph.D. Thesis, Linköping, Sweden, Oct. 2025.
- Mody SY, *Enhanced Receiver Design for Ultra-Reliable Low-Latency Short Data Communications*, Ph.D. Thesis, Eurecom, France, Oct. 2024.
- Borja Rodriguez Galvez, *An Information-Theoretic Approach to Generalization Theory*, Ph.D. Thesis, KTH, Sweden, Apr. 2024.
- Juan Vidal Alegría, *Beyond Massive MIMO: Trade-offs and Opportunities with Large Multi-Antenna Systems*, Ph.D. Thesis, Lund University, Sweden, June 2023
- Antonino Favano, *The Capacity of Amplitude-Constrained Vector Gaussian Channels*, Ph.D. Thesis, Politecnico di Milano, Italy, Jan. 2023
- Linghui Zhou, *Information-Theoretic Privacy and Secrecy in Biometric Identification and Authentication*, Ph.D. Thesis, Royal Institute of Technology, Sweden, Oct. 2022.
- Kassem Saied, *Quasi-Cyclic Short Packet Transmission for IoT*, Ph.D. Thesis, Université Bretagne Sud, France, Mar. 2022.
- Emmanouil Foutoulakis, *Performance and Optimization Aspects of Time Critical Networking*, Ph.D. Thesis, Linköping University, Sweden, Oct. 2021.
- Khac-Hoang Ngo, *Non-coherent Wireless Communications: Fundamental Limits and System Design*, Ph.D. Thesis, University of Paris-Saclay, France, July 2020
- Apostolos Avranas, *Resource Allocation for Latency Sensitive Wireless Systems*, Ph.D. Thesis, Telecom Paris, France, Jun. 2020.
- Emmanouil Fountoulakis, *Performance Analysis and Optimization for Time Critical Networking*, Licentiate Thesis, Linköping University, Sweden, Oct. 2019.
- Jun Li, *Ultra-low latency communication for 5G transport network*, PhD Thesis, Royal Institute of Technology, Sweden, Sep. 2019.
- Sebastian Schiessl, *Performance Trade-offs for Ultra-Reliable Low-Latency Communication Systems*, PhD Thesis, Royal Institute of Technology, Sweden, Jun. 2019.
- Silvio Mandelli, *Analysis of Wiener phase noise issues in optical transmission systems*, PhD Thesis, Politecnico di Milano, Italy, Jan. 2016.

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- Simone Pecorino, *Information rates of channels with memory by Bayesian tracking*, PhD Thesis, Politecnico di Milano, Italy, Jan. 2016.
- Zhao Wang, *Interference Alignment with Imperfect Channel Knowledge and Secrecy Constraints*, PhD Thesis, Royal Institute of Technology, Sweden, May 2015.
- Günther Koliander, *Information-theoretic analysis of noncoherent block-fading channels and singular random variables*, PhD Thesis, Vienna University of Technology, Vienna, Austria, Apr. 2015
- Hien Quoc Ngo, *Massive MIMO: fundamentals and system designs*, PhD Thesis, Linköping University, Sweden, Mar. 2015
- David Gustafsson, *Extending the bandwidth of the Doherty power amplifier*, PhD thesis, Chalmers University of Technology, Sweden, Sep. 2014
- Nicolas Schrammar, *On deterministic models for Gaussian networks*, PhD thesis, KTH Royal Institute of Technology, Sweden, Jun. 2013
- Hien Quoc Ngo, *Performance bounds for very large multiuser MIMO systems*, Licentiate thesis, Linköping University, Sweden, Dec. 2012
- Lu Lu, *Wireless broadcasting with network coding*, Licentiate thesis, KTH Royal Institute of Technology, Sweden, Sep. 2011
- P. Kuppinger, *General uncertainty relations and sparse signal recovery*, PhD thesis, ETH Zurich, Switzerland Feb. 2011

FUNDING

- ◇ “Wireless beyond standardization”, Swedish Research Council 2025-07508, network grant for planning future excellent clusters for groundbreaking technologies, 1 200 kSEK, responsibility: **co-investigator**
- ◇ “Understanding transformers via compression and discrimination”. Wallenberg AI, autonomous systems, and software program, 2026–2029, academic doctoral student project, responsibility: **principal investigator**
- ◇ “R-SAFE—runtime safe monitors for automotive ML-based functions”, Wallenberg AI, autonomous systems, and software program, 2025–2028 industrial doctoral student project, (Volvo Car Corporation), responsibility: **co-investigator**
- ◇ “Goal-oriented communications: When is semantic feedback useful?”, Swedish Research Council 2022-05571 project grant, 3 600 kSEK, responsibility: **co-investigator**
- ◇ “SAICOM: Software Artificial Intelligence for Communications”, Swedish Foundation for Strategic Research (SSF), Future software systems, 2022–2026, 35 000 kSEK, responsibility: **co-investigator**
- ◇ “Evaluated mutual information bounds for learning and metalearning”, Wallenberg AI, autonomous systems, and software program 2023–2027, academic doctoral student project, responsibility: **principal investigator**
- ◇ “Energy-efficient massive random access for real-time distributed autonomous systems”, Swedish Research Council 2021-04970 project grant, 2022–2025, 4 000 kSEK, responsibility: **principal investigator**
- ◇ “LANTERN: Low-latency and private edge computing in random-access networks”, Marie Skłodowska-Curie Individual Fellowship 2021–2022, responsibility: **co-supervisor** (fellow: K.-H. Ngo)
- ◇ “Information-theoretic limits for deep neural networks”, Marie Skłodowska-Curie Individual Fellowship 2020–2021, responsibility: **supervisor** (fellow: A. Rezazadeh)
- ◇ “Generalization bounds for deep neural networks: Insight and Design” Wallenberg AI, autonomous systems, and software program, academic doctoral student project, 2020–2025, responsibility: **co-investigator**

- ◇ “Non-coherent low-latency communications”, German DAAD-DLR fellowship, 25 kSEK, 2019, responsibility: **principal investigator**
- ◇ “Distributed massive MIMO with low-precision components”, Swedish Foundation for Strategic Research (SSF), Industrial PhD grant, 2 500 kSEK, 2020–2024, responsibility: **principal investigator**
- ◇ “Secure, Private, and Low-Latency Cloud Connectivity for IoT Applications”, WASP expedition project, 2019–2021, responsibility: **principal investigator**
- ◇ “Communication in time: theory and practice for the automated society”, Swedish Research Council, *2018-04359 project grant* 2019–2022, 4 000 kSEK, responsibility: **co-investigator**
- ◇ “Low-Latency Wireless Random Access for IoT connectivity”, Swedish Foundation for Strategic Research (SSF), Mobility Grant, 933 kSEK, 2018–2019, responsibility: **principal investigator**
- ◇ “Protection in a Power-Electronic Converter Dominated Medium-Voltage Microgrid Utilizing Digital Communication”, Chalmers Energy Area of Advance, 2MSEK, 2018–2019, responsibility: **co-investigator**
- ◇ “Wireless communication for stability and performance enhancement in converter-dominated AC microgrids”, Chalmers E2 seed fund, 250 kSEK, 2018, responsibility: **co-investigator**
- ◇ “INNER–information theory of deep neural networks”, Chalmers E2 seed fund, 500 kSEK, 2018, responsibility: **principal investigator**
- ◇ “SWIFT–short-packet wireless information theory”, Swedish Research Council, *2016-03293 project grant* 2017–2020, 3 680 kSEK, responsibility: **principal investigator**
- ◇ “mmMAGIC–Millimetre-Wave Based Mobile Radio Access Network for Fifth Generation Integrated Communications”, European Commission, Horizon 2020, 2015–2017, responsibility: **project member**
- ◇ “Massive MIMO systems with low-precision converters”, Swedish Foundation for Strategic Research (SSF), Industrial PhD grant, 2 500 kSEK, 2015–2019, responsibility: **principal investigator**
- ◇ “Theory and practice for optimum spectral efficiency for ad-hoc wireless networks with strict requirements on latency and reliability”, Swedish Research Council, *2014-6066 project grant* 2015–2018, 3 643 kSEK, responsibility: **co-investigator**
- ◇ “Multi-antenna technologies for wireless access and backhaul (MATWAB)”, VINNOVA, 3570 kSEK, 2015–2016; responsibility: **co-investigator and project manager**
- ◇ “Massive MIMO: from hype to roll-out”, Ericsson AB, 4000 kSEK, 2014–2016, **co-investigator**
- ◇ “Nonlinear hardware impairments in massive MIMO systems”, Swedish Foundation for Strategic Research (SSF), Strategic Mobility Grant, 942 kSEK, 2014–2015, responsibility: **principal investigator**
- ◇ “Optical interference is not noise”, Swedish Research Council, *2013-5271 project grant*, 4 720 kSEK, 2014–2017; responsibility: **co-investigator**
- ◇ “Multipath test environment for future communication systems” Swedish Research Council, *2013-5718 industrial Ph.D. project grant*, 2 156 kSEK, 2014–2017; responsibility: **co-investigator**
- ◇ “Fundamental limits of user cooperation in wireless networks”, Swedish Research Council, *2012-4571 Junior researcher grant*, 3 280 kSEK; 2013–2016; responsibility: **principal investigator**
- ◇ “Handling the data flood in wireless communications”, Chalmers *Area of Advance* initiative: Information and Communication Technology; 8 000 kSEK; 2010–2014; responsibility: **principal investigator**

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PUBLICATIONS: OVERVIEW Since 2001, Dr. Durisi has published a book and a monograph, 66 journal papers, and 119 conference papers. These papers received 11 778 citations according to [Google Scholar](#), with a corresponding *h-index* of 49. The publications are listed below in reverse chronological order.

PREPRINTS

- [P2] A. Bordbar, A. Hu, and G. Durisi, “Distributed massive MIMO with 1-bit radio-over-fiber fronthaul: Uplink spectral efficiency and power control,” June 2026.
- [P1] G. Durisi and A. Lancho, *Transmitting short packets over wireless channels—an information-theoretic perspective*, Lecture Notes, Nov. 2020. [[Online](#)]

PUBLICATIONS: BOOKS AND MONOGRAPHS

- [B1] A. Zaidi, F. Athley, J. Medbo, U. Gustavsson, G. Durisi, and X. Cheng, *5G Physical Layer: principles, models and technology components*. Academic Press, 2018.
- [B2] F. Hellström, G. Durisi, B. Guedj, and M. Raginsky, *Generalization Bounds: Perspectives from Information Theory and PAC-Bayes*, vol. 18, pp. 1–223, Foundations and Trends in Machine Learning, Jan. 2025

PUBLICATIONS: BOOK CHAPTERS

- [BC1] G. Durisi, G. Liva, and Y. Polyanskiy, “Short-packet transmission,” in *Information Theoretic Perspectives on 5G Systems and Beyond*, I. Marić, S. Shamai (Shitz), and O. Simeone, Eds. London, U.K.: Cambridge Univ. Press, Jul. 2020.
- [BC2] G. Durisi, U. G. Schuster, V. I. Morgenshtern, H. Bölcskei, and S. Shamai (Shitz), “Information theory of underspread WSSUS channels,” in *Wireless Communications over Rapidly Time-Varying Channels*, F. Hlawatsch and G. Matz, Eds. Academic Press, Mar. 2011.

PUBLICATIONS: JOURNAL PAPERS

- [J66] J.-F. Chamberland, K. Narayanan, K.-H. Ngo, G. Durisi, V. T. Kande, and C. S. K. Valmeekam, “Prediction-based compression using large language models,” *IEEE BITS*, 2026, to appear
- [J65] K. Okumus, K.-H. Ngo, G. Durisi, and E. G. Ström, “Type-based unsourced multiple access over fading channels in distributed MIMO with application to multi-target localization,” *IEEE Trans. Wireless Commun.*, 2026, to appear.
- [J64] M. Zecchin, U. K. Ganesan, G. Durisi, P. Popovski, and O. Simeone, “Prediction-powered communication with distortion guarantees,” *IEEE J. Sel. Areas Info. Theory*, Feb. 2026, to appear.
- [J63] L. Aabel, G. Durisi, F. Olofsson, E. Börjeson, M. Coldrey, and C. Fager, “Insights on the uplink operation of a 1-bit radio-over-fiber architecture in multi-user D-MIMO communication,” *IEEE Trans. Microw. Theory Tech.*, 2025, to appear.
- [J62] K.-H. Ngo, G. Durisi, A. Munari, F. Lazaro, and A. Graell i Amat, “Timely status updates in slotted ALOHA networks with energy harvesting,” *IEEE Trans. Commun.*, vol. 73, pp. 7288–7303, Sept. 2025.
- [J61] A. Sauter, A. O. Kislal, G. Durisi, G. Liva, B. Matuz, and E. G. Ström, “Undetected error probability in the short blocklength regime: Approaching finite-blocklength bounds with polar codes,” *IEEE Trans. Commun.*, vol. 73, pp. 7007–7020, Sept. 2025.
- [J60] H. Guo, H. Wymeersch, B. Makki, H. Chen, Y. Wu, G. Durisi, M. F. Keskin, M. M. Moghaddam, C. Madapatha, H. Yu, P. Hammarberg, H. Kim, and T. Svensson, “Integrated communication, localization and sensing in 6G D-MIMO networks,” *IEEE Wireless Commun. Mag.*, vol. 32, pp. 214–221, Feb. 2025.

- [J59] A. O. Kislal, M. Rajiv, G. Durisi, E. G. Ström, and U. Mitra, “Is synchronization a bottleneck for pilot-assisted URLLC links?,” *IEEE Trans. Wireless Commun.*, vol. 23, pp. 17945–17958, Dec. 2024.
- [J58] A. E. Kalør, G. Durisi, S. Coleri, S. Parkvall, W. Yu, A. Mueller, and P. Popovski, “Wireless 6G connectivity for massive number of devices and critical services,” *Proc. IEEE*, pp. 1–23, Nov. 2024.
- [J57] A. Hu, L. Aabel, G. Durisi, S. Jacobsson, M. Coldrey, and C. Fager, “EVM analysis of distributed massive MIMO with 1-bit radio-over-fiber fronthaul,” *IEEE Trans. Commun.*, vol. 72, pp. 7342–7356, Nov. 2024.
- [J56] L. Aabel, S. Jacobsson, M. Coldrey, F. Olofsson, G. Durisi, and C. Fager, “A TDD distributed MIMO testbed using a 1-bit radio-over-fiber fronthaul architecture,” *IEEE Trans. Microw. Theory Tech.*, vol. 72, pp. 6140–6152, Oct. 2024.
- [J55] K.-H. Ngo, G. Durisi, A. Graell i Amat, P. Popovski, A. E. Kalør, and B. Soret, “Un-sourced multiple access with common alarm messages: Network slicing for massive and critical IoT,” *IEEE Trans. Commun.*, vol. 72, no. 2, pp. 907–923, Feb. 2024.
- [J54] A. Munari, F. Lazaro, G. Durisi, and G. Liva, “The dynamic behavior of frameless ALOHA: Stability, throughput, and age of information,” *IEEE Trans. Commun.*, vol. 71, no. 12, pp. 6914–6927, Dec. 2023.
- [J53] Y. Ettetfagh, S. R. Aghdam, G. Durisi, S. Jacobsson, M. Coldrey, and C. Studer, “Performance of quantized massive MIMO with fronthaul rate constraint over quasi-static channels,” *IEEE Access*, vol. 11, 2023.
- [J52] A. Lancho, G. Durisi, and L. Sanguinetti, “Cell-free massive MIMO for URLLC: A finite-blocklength analysis,” *IEEE Trans. Wireless Commun.*, vol. 22, no. 12, pp. 8723–8735, Dec. 2023.
- [J51] A. O. Kislal, A. Lancho, G. Durisi, and E. Ström, “Efficient evaluation of the error probability for pilot-assisted URLLC with Massive MIMO,” *IEEE J. Sel. Areas Commun.*, vol. 41, no. 7, pp. 1969–1981, July 2023.
- [J50] K.-H. Ngo, A. Lancho, G. Durisi, and A. Graell i Amat, “Un-sourced multiple access with random user activity,” *IEEE Trans. Inf. Theory*, vol. 69, no. 7, pp. 4537–4558, July 2023.
- [J49] S. T. Jose, O. Simeone, and G. Durisi, “Transfer meta-learning: Information-theoretic bounds and information meta-risk minimization,” *IEEE Trans. Inf. Theory*, vol. 68, no. 1, pp. 474–501, Jan. 2022.
- [J48] F. Hellström and G. Durisi, “Corrections to ‘Generalization bounds via information and conditional information density’,” *IEEE J. Sel. Areas Inf. Theory*, vol. 2, no. 3, pp. 1072–1073, Sep. 2021.
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SOFTWARE

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TUTORIALS

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INVITED TALKS

- [I42] “Prediction-aided sequential communication of individual sequences with distortion guarantees”, Workshop on Information Theory for Future Networks, Mar. 2026
- [I41] “Theoretical performance guarantees for partial domain adaptation via partial optimal transport”, Wireless AI and Data Technology Workshop Stockholm, Sweden, Feb. 2026
- [I40] “Theoretical performance guarantees for partial domain adaptation via partial optimal transport”, Workshop in Advancements in High-Dimensional Methods for Machine Learning, Gothenburg, Sweden, May 2025
- [I39] “Packet and error detection for short-packet transmission”, London Symposium on Information Theory, Cambridge, United Kingdom, May 2025
- [I38] “Type-based unsourced random access”, International ITG Conference on Systems, Communications, and Coding, Karlsruhe, Germany, March 2025.
- [I37] “Generalization bounds: Perspectives from PAC-Bayes and information theory”, Aalborg, Denmark, April 2024.
- [I36] “Short packets over a massive random-access channel”, London Symposium on Information Theory, London, U.K., May 2023
- [I35] “Seeking information-theoretic bounds that explain generalization”, Information theory and tapas workshop, Madrid, Spain, Jan. 2023
- [I34] “Short packets over wireless fading networks”, Digital Futures Seminar, Stockholm, Sweden, Oct. 2022
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- [I32] “Short packets over wireless fading networks”, Keynote talk, Workshop on short packet communications for 6G mission-critical applications, International Conference on Communications (ICC), May 2022
- [I31] “Short packets over wireless fading networks”, Plenary talk, Information Theory Workshop (ITW), Sep. 2021

- [I30] “Fast-Rate Loss Bounds via Conditional Information Measures with Applications to Neural Networks”, London Symposium on Information Theory, May 2021.
- [I29] “Short-Packet Transmission via Variable-Length Codes in the Presence of Noisy Stop Feedback”, DLR-MIT-TUM Workshop on Coding and Random Access, Munich, Germany, Feb. 2020.
- [I28] “Playing with bits”, Lunch talks at Ericsson Series, Gothenburg, Sweden, Sep. 2019.
- [I27] “Fundamentals of short-packet transmission”, Plenary talk at the Small-Data Networks Special Session, Int. Symp. Personal, Indoor and Mobile Radio Commun., Bologna, Italy, Sep. 2018
- [I26] “Delay and peak-age violation probability in short-packet transmissions”, Communication Theory Workshop (CTW), Miramar Beach, Florida, May 2018
- [I25] “Short-packet over fading channels: pilot-assisted or noncoherent transmission?”, NSF Workshop on Low-Latency Wireless Random-Access, MIT, Cambridge, MA, USA, Nov. 2017
- [I24] “Short-packet transmission”, tutorial presentation at “ITG-Fachgruppe”: applied information theory, Bremen, Germany, Oct. 2017
- [I23] “Quantized precoding for massive MU-MIMO”, Ericsson, Stockholm, Sweden, Nov. 2016
- [I22] “Latency-reliability trade-off in short packet communications”, Communication Theory Workshop (CTW), Nafplio, Greece, May 2016
- [I21] “My path through information theory: From double asymptotics to finite blocklength”, Claude Elwood Shannon 100th Birthday Celebration, Paderborn, Germany, May 2016
- [I20] “How much energy is needed to transmit k bits over fading channels?”, Technische Universität München (TUM), Dec. 2015
- [I19] “Towards low-latency wireless communications: the art of sending short packets”, GRD ISIS workshop on recent advances in network information theory and coding theory, Lyon, Nov. 2015
- [I18] “Short packet communications with multiple antennas”, Vienna University of Technology, Apr. 2015
- [I17] “Short packet communications with multiple antennas”, Nordic Workshop on System & Network Optimization for Wireless, Geilo, Norway, Mar. 2015
- [I16] “Short packet communications with multiple antennas”, Linköping University, Sweden, Mar. 2015
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- [I14] “Massive MIMO uplink with 1 bit ADC”, Ericsson AB, Stockholm, Sweden, Nov., 2014
- [I13] “On the cost of CSI acquisition in large MIMO systems”, Communication Theory Workshop, Phuket, Thailand, Jun., 2013
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- [I7] “Degrees of freedom of multiple-antenna systems under channel uncertainty”, Politecnico di Torino, Turin, Italy, Sep. 2011

Giuseppe Durisi: Curriculum Vitae

- [I6] “Degrees of freedom of multiple-antenna systems under channel uncertainty”, KTH Royal Institute of Technology, Stockholm, Sweden, Sep. 2011
- [I5] “Compressed sensing and beyond”, Gothenburg University, Sweden, Aug. 2011
- [I4] “Handling the data flood in wireless communications”, Chalmers University of Technology, Gothenburg, Sweden, May 2010.
- [I3] “On the noncoherent capacity of wireless channels of practical interest”, Joint UMIC and ISS seminar, RWTH Aachen, Germany, Jun. 2009.
- [I2] “On the capacity of real-world fading channels”, ITG meeting—Algorithms for signal processing, Zurich, Switzerland, Mar. 2009.
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